Mortgage-Backed Securities Prepayment Prediction Using Machine Learning

Project Report

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Abstract

This report presents a supervised machine-learning study that predicts whether a mortgage loan will be prepaid, using the Freddie Mac 2024 sample of 50,000 loans. Four classifiers (Logistic Regression, RBF Support Vector Machine, Random Forest and a feed-forward neural network) are trained under a common preprocessing pipeline with class-imbalance handling. Decision thresholds and the final model are chosen using a separate validation set only. The RBF SVM is selected (validation F1 = 0.4304) and attains an F1-score of 0.4432 and a ROC-AUC of 0.7393 on the unseen test set.

1. Introduction

Mortgage-backed securities (MBS) derive their cash flows from pools of underlying home loans. When borrowers repay a loan earlier than scheduled, the timing of those cash flows changes, which affects the value of the security and creates prepayment risk for investors. Anticipating which loans are likely to be prepaid is therefore useful for understanding and managing this risk.

The aim of this project is to build a binary classification system that, given the characteristics of a mortgage at origination, predicts whether the loan is prepaid (class 1) or not (class 0). Because prepaid loans form a minority of the data, model quality is judged primarily by F1-score rather than accuracy alone.

The main contributions of the work are: (i) construction of a loan-level prepayment target from the Freddie Mac origination and performance files; (ii) a like-for-like comparison of four model families under identical preprocessing; (iii) explicit handling of class imbalance through balanced class weights and F1-oriented threshold tuning; and (iv) a leakage-safe selection protocol in which the test set is touched only once, after the model and threshold have been fixed.

The remainder of the report is organised as follows. Section 2 describes the data and target, Section 3 the features and preprocessing, Section 4 the models and evaluation protocol, Section 5 the results, and Sections 6 to 9 the discussion, limitations, future scope and conclusion.

2. Dataset and Target Construction

2.1 Data sources

The study uses the Freddie Mac 2024 sample, which consists of two files (Table 1). The origination file records the characteristics of each loan at the time it was issued, while the performance file contains month-by-month records of the loan's status. The two files are linked through the loan sequence number.

Table 1: Freddie Mac 2024 sample files.

File	Records	Columns	Content
Origination	50,000	31	Loan characteristics at issuance
Performance	952,865	35	Monthly loan status and balance
Modelling dataset	50,000	32	Origination fields plus target

2.2 Target definition

A loan is labelled as prepaid if any of its performance records has a zero-balance code of 1. Performance records are grouped by loan sequence number and the maximum of this indicator gives the loan-level target, so that a single prepaid observation marks the loan as prepaid. Joining the target to the origination data yields the 50,000-loan modelling dataset. The resulting class distribution is imbalanced (Table 2).

Table 2: Target distribution of the modelling dataset.

Class	Loans	Share (%)
Not prepaid (0)	40,823	81.65
Prepaid (1)	9,177	18.35

2.3 Why class imbalance matters

With roughly 82% of loans in the majority class, a model could obtain a high accuracy simply by favouring the 'not prepaid' label. Accuracy alone therefore cannot indicate whether the minority class is being identified. For this reason the project uses balanced class weights where the algorithm supports them and emphasises F1-score, which balances precision and recall.

3. Features and Preprocessing

3.1 Feature set

Twenty-eight origination-time features are used, 14 numerical and 14 categorical (Table 3). Loan identifiers are excluded, and only information available when the loan is issued is used as input.

Table 3: Input features used by the models.

Type	Count	Features
Numerical	14	credit score, first payment date, maturity date, MSA, mortgage insurance %, number of units, original CLTV, DTI, UPB, LTV, interest rate, postal code, loan term, number of borrowers
Categorical	14	first-time homebuyer flag, occupancy status, channel, prepayment penalty flag, amortization type, property state, property type, loan purpose, seller, servicer, special eligibility program, property valuation method, relief refinance indicator, interest-only indicator

3.2 Preprocessing pipeline

Numerical features are imputed with the median and standardised. Categorical features are imputed with the most frequent value and one-hot encoded, with unseen categories ignored at prediction time. A column transformer applies both branches together, and the transformer is combined with each classifier in a single scikit-learn pipeline.

Using a pipeline has two benefits. First, preprocessing is fitted on the training data only, so no information from the validation or test sets leaks into the transformations. Second, exactly the same transformation is reapplied automatically during validation, testing and later deployment.

4. Methodology

4.1 Data split

The data are divided by stratified sampling into training, validation and test sets (Table 4), which preserves the class proportions in each subset. A fixed random state of 42 makes the split reproducible.

Table 4: Train, validation and test split.

Subset	Loans	Share	Purpose
Training	30,000	60%	Fit model parameters
Validation	10,000	20%	Tune thresholds and select the model
Test	10,000	20%	Final evaluation only

4.2 Models

Four classifiers are compared; their configurations are listed in Table 5.

Logistic Regression is a linear baseline with an interpretable decision boundary. It maps a linear score to a probability through the sigmoid function. **The RBF Support Vector Machine** seeks a large-margin decision boundary and, through the RBF kernel, can represent non-linear boundaries. Probability calibration is deliberately

switched off because it is very slow; the decision-function scores are used instead for threshold tuning and ROC-AUC.

The Random Forest is an ensemble of decision trees that captures non-linear relationships and feature interactions and reduces dependence on any single tree. **The feed-forward neural network** is a multilayer perceptron with hidden layers of 64 and 32 units, ReLU activations and the Adam optimiser; early stopping controls over-fitting and unnecessary training.

Table 5: Model configurations.

Model	Configuration
Logistic Regression	max_iter = 1000, balanced class weights
RBF SVM	C = 1.0, gamma = scale, balanced class weights, probability = False
Random Forest	200 trees, max_depth = 18, min_samples_leaf = 3, balanced class weights
Neural Network (MLP)	hidden layers (64, 32), ReLU, Adam, early stopping, max_iter = 100

4.3 Threshold optimisation

A model produces a score that must be converted into class 0 or 1 by a threshold, and a default of 0.5 is not assumed here. For each model, candidate thresholds are evaluated on the validation data and the threshold giving the highest validation F1-score is selected. When a model produces many distinct scores, about 500 quantile-based candidates are searched. The test set is never used to choose a threshold.

4.4 Leakage-safe model selection

The selection procedure follows six steps: (1) all four models are trained on the training set; (2) each threshold is tuned on the validation data; (3) the validation F1-scores are compared; (4) the best model and its threshold are fixed; (5) that model and threshold are applied once to the unseen test set; and (6) test metrics are reported only after selection is complete. The complete workflow is shown in Figure 1.

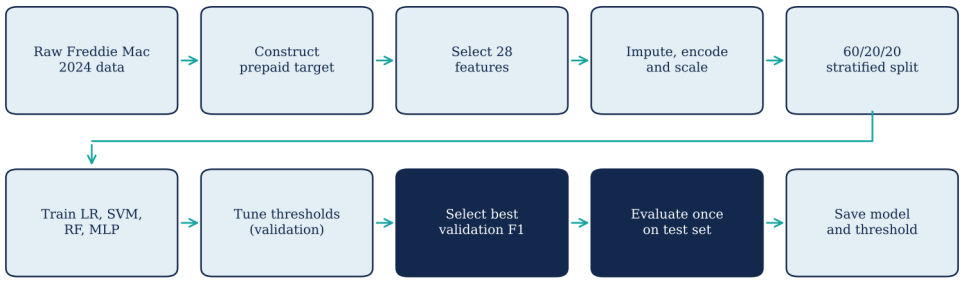


Figure 1: End-to-end workflow of the project.

5. Results

5.1 Model selection and test-set results

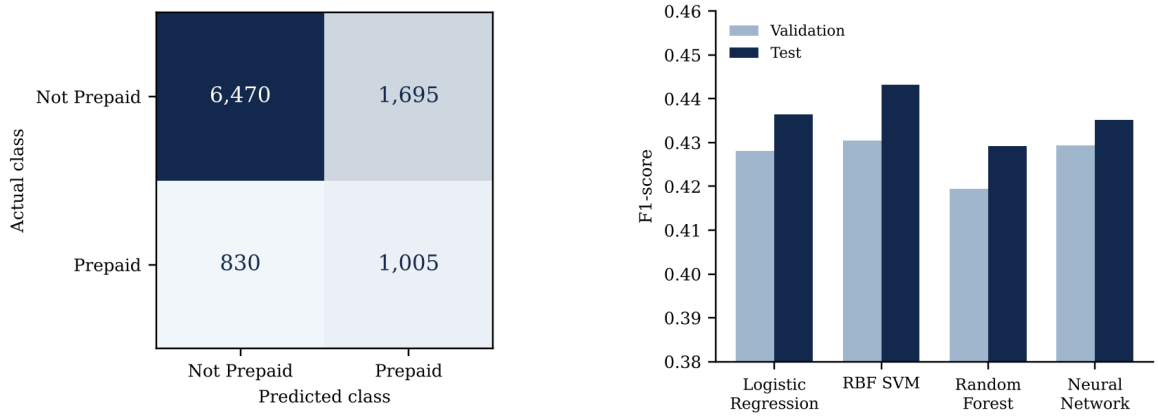
On the validation set the RBF SVM obtained the highest F1-score (0.4304), followed closely by the neural network (0.4294), Logistic Regression (0.4281) and the Random Forest (0.4194). The RBF SVM was therefore selected with a threshold of 0.3167. This choice reflects the validation evidence for this dataset and not a claim that the SVM is always the best algorithm.

Table 6 gives the validation F1 together with the performance of all four models on the same unseen test set, using the thresholds fixed on the validation data. The test figures are for reporting only: the winner had already been chosen using validation F1.

Table 6: Validation F1 and test-set performance of all models.

Model	Threshold	Val. F1	Accuracy	Precision	Recall	Test F1	ROC-AUC
Logistic Regression	0.5461	0.4281	0.7095	0.3389	0.6131	0.4365	0.7375
RBF SVM (selected)	**0.3167**	**0.4304**	**0.7475**	**0.3722**	**0.5477**	**0.4432**	**0.7393**
Random Forest	0.4762	0.4194	0.7529	0.3725	0.5063	0.4292	0.7345
Neural Network	0.2230	0.4294	0.7327	0.3554	0.5613	0.4352	0.7349

The selected RBF SVM achieves a test accuracy of 74.75%, precision of 37.22%, recall of 54.77%, F1-score of 44.32% and ROC-AUC of 0.7393. Figure 2(a) shows its confusion matrix and Figure 2(b) compares validation and test F1 across the models.



(a) Test confusion matrix, RBF SVM *(b) F1-score by model*

Figure 2: Performance of the final model and comparison across models.

5.3 Error analysis of the final model

The confusion matrix of the final model on the 10,000 test loans contains 6,470 true negatives, 1,695 false positives, 830 false negatives and 1,005 true positives. Table 7 summarises the quantities that follow from it.

Table 7: Quantities derived from the test confusion matrix (RBF SVM).

Quantity	Value	Meaning
True positives	1,005	Prepaid loans correctly flagged
False negatives	830	Prepaid loans missed
False positives	1,695	Non-prepaid loans flagged as prepaid
Recall	54.77%	1,005 of 1,835 actual prepaid loans
Precision	37.22%	1,005 of 2,700 loans flagged
Specificity	79.24%	6,470 of 8,165 non-prepaid loans

6. Discussion

6.1 Comparison of the models

The four models perform very similarly: test F1 lies between 0.429 and 0.443 and ROC-AUC between 0.735 and 0.739. The differences among the top models on validation F1 are small, so the choice of the RBF SVM rests on a narrow margin. This suggests that the origination features carry a moderate signal that is largely shared by all four models, and that more complex models do not extract much more of it.

The Random Forest has the highest test accuracy (75.29%) but was not selected, because selection is based on validation F1. Logistic Regression has the highest test recall (61.31%) but lower precision and F1 than the SVM. The RBF SVM has the highest test F1 (44.32%) among the four models.

6.2 Accuracy versus F1-score

Accuracy alone is misleading for this problem. A classifier that always predicts the majority class would reach about 81.65% accuracy on the test set while detecting no prepayments, which is higher than the selected model's 74.75%. The F1-score, precision, recall and ROC-AUC give a more meaningful picture of how well the minority class is identified.

6.3 Precision and recall trade-off

Optimising F1 gives a model with recall above 50% but modest precision of about 37%: it finds more than half of the prepaid loans but also flags a considerable number of loans that are not prepaid. This is a consequence of the threshold of 0.3167, which favours recall. If missed prepayments were costlier than false alarms, a lower threshold could be chosen; in the opposite case a higher one would suit better. The threshold is therefore a business decision as much as a statistical one.

6.4 Model saving and verification

After selection, the final pipeline, its threshold and a metadata record (model name, threshold, dataset and selection metric) were saved (Table 8). The notebook then reloads them and recomputes the test F1, reproducing exactly 0.4432, which confirms that the saved artefacts give the same result as the original run and are ready for use in a Streamlit application.

Table 8: Saved artefacts.

File	Content
mortgage_prepayment_final.pkl	Final trained pipeline (preprocessing and RBF SVM)
final_threshold.pkl	Selected threshold (0.3167)
final_model_metadata.pkl	Model name, threshold, dataset and selection metric

6.5 Relation to the reference paper

This work should be described as an extension of the reference paper rather than a direct accuracy comparison, since the dataset period, sample, preprocessing, target construction and experimental set-up differ. The additions are the newer 2024 sample, the Random Forest in the model comparison, explicit handling of class imbalance with balanced class weights, validation-based threshold tuning, leakage-safe model selection and deployment-ready model saving.

7. Limitations

The experiment uses a sample of 50,000 loans from a single vintage, so the findings may not carry over to other periods or to the full population. The feature set is based mainly on origination characteristics and omits information that develops later in a loan's life. The validation F1-scores of the top models are close, so the margin between them is small and could change with a different split. Finally, the target is a binary outcome and does not model when a prepayment occurs, so loans that have not yet had time to prepay are treated as non-prepaid.

8. Future Scope

Several extensions are natural. Richer temporal and macroeconomic variables, such as the gap between prevailing market rates and the loan rate, could be added. Temporal generalisation can be evaluated on later loan vintages. Hyperparameters can be optimised systematically rather than set manually, calibrated probabilities can be investigated where they are needed, and a time-to-event (survival) formulation can replace the binary target. The saved model can then be deployed through the Streamlit application.

9. Conclusion

An end-to-end mortgage prepayment classification pipeline was built using the Freddie Mac 2024 sample, and four machine-learning models were compared under a common preprocessing and evaluation framework. Validation F1 was used for threshold optimisation and model selection, and the RBF SVM was chosen with a threshold of 0.3167 and a validation F1 of 0.4304. On the unseen test set it achieved an F1-score of 0.4432 and a ROC-AUC of 0.7393. The model, its threshold and its metadata have been saved and verified for deployment.